

The Shape of the Tax

Transaction Shifting at New York City's 2019 Transfer-Tax Notches and the Mortgage Recording Tax

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Abstract

New York's 2019 reform introduced seven buyer-paid transfer-tax notches. At the three well-populated thresholds, recorded sales shifted below the threshold relative to the same prices before reform and to never-taxed round prices. The unchanged \$1 million notch supplies an additional comparison. Cross-threshold responses reject a proportional benchmark, but differences between markets prevent interpreting this as a rate experiment. Conditional on cross-threshold relocation, implied avoided tax is \$2–3 million annually. Loan records also support a gross mortgage-recording-charge benchmark and static costing of continuous replacements. Neither exercise identifies economic incidence or an optimal tax.

JEL: H23, H71, K34, R31, R38. *Keywords:* transfer taxes; mortgage recording tax; statutory liability; notches; bunching; housing transactions.

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1 Introduction

Housing transaction taxes are often notches: the rate applies to the whole price once a threshold is crossed, so one more dollar of price can carry thousands of dollars of tax. New York’s 2019 budget created seven such notches at once. A buyer-paid supplemental tax took effect on 1 July 2019 at prices from \$2,000,000 to \$25,000,000, while the \$1,000,000 mansion tax, in place since 1989, did not change. The same thresholds can therefore be observed before and after they were taxed, next to an always-treated control, and the seven new notches share a start date while their dollar jumps range from \$5,000 to \$100,000. This paper uses that reform to ask what a multi-threshold, buyer-paid notch schedule does to transactions, and what that implies for the shape of housing transaction taxes.

1.1 What this paper does

First, it uses the 2019 reform to measure transaction shifting at tax thresholds.

New York City has nine residential transfer-tax notches. Seven were created effective 1 July 2019 by a buyer-paid supplemental tax whose rate steps up at \$2,000,000, \$3,000,000, \$5,000,000, \$10,000,000, \$15,000,000, \$20,000,000 and \$25,000,000 and applies to the whole price; at \$3,000,000 the seller’s base rate also rises. The mansion tax at \$1,000,000 has been 1% since 1989. The years before July 2019 are therefore a placebo at the same price points, and the \$1,000,000 threshold is an always-treated control. After the reform, transactions shift around the new thresholds that have enough sales to measure, while the change at the always-treated control is small. Each new threshold’s response is larger than the same statistic at all sixteen never-taxed round prices whose estimation windows contain no threshold, and a year-by-year event study shows the shift starting with the reform, with no break at a fake reform date in 2018. The response appears among houses and condominiums and in Manhattan and the other boroughs, and removing foreclosure and lender deeds does not change it. At \$2,000,000 and \$3,000,000 it empties a range of prices two to three times as wide as the tax jump. Shifting remains evident without company sellers; among company-seller sales, a proxy that includes sponsor sales whose prices developers set in advance, it is clear only at \$3,000,000. Under a cross-threshold relocation assumption, estimated avoidance is \$2–3 million a year, 2–3% of schedule-implied revenue between \$2,000,000 and \$6,000,000. A simple count shows the response without any estimator. Before the reform, \$2,000,000 was an ordinary focal price, with 7.8 times the surrounding density of sales at exactly that price, and four sales were recorded one dollar below it between January 2016 and March 2019. After the reform the spike falls to 1.6 times the surrounding density, and 131 sales were recorded one dollar below. At \$5,000,000 no sale was recorded one dollar below the

threshold before the reform, and 26 have been since.

The estimated response does not rise in proportion to the buyer’s rate increment. The point estimates at \$5,000,000, where the buyer’s jump is \$37,500, and \$2,000,000, where it is \$5,000, are nearly the same. The sampling bootstrap rejects the proportional benchmark; comparisons with placebo noise give weaker, reference-dependent evidence against it. The interval on their difference is wide and does not establish equal responses. The design does not identify the effect of statutory payer identity. Adding a seller-paid component at \$3,000,000 does not significantly enlarge the response, and the city’s seller-paid threshold cannot serve as a comparison, because its higher rate starts *above* \$500,000: a sale at exactly \$500,000 is itself taxed at the lower rate. Above \$5,000,000 the market is thin, but the simplest count registers responses at \$10,000,000 and \$15,000,000 whose point estimates are no larger than at \$2,000,000.

Second, it applies the result to the mortgage recording tax. New York City’s mortgage recording tax charges purchase borrowing. The borrower-side statutory rate is 1.80% on loans below \$500,000 and 1.925% at or above. On a \$900,000 condominium with a \$720,000 mortgage, the schedule assigns \$13,860 to the borrower, while a purchase without a recorded mortgage pays no purchase-mortgage recording tax. Matching New York City Department of Finance sales records to the city’s land-records system (ACRIS) recovers the loan amount alongside the price for 172,725 financed purchases in the four boroughs ACRIS covers: 125,023 houses and 47,702 condominiums. Applying the published recording-tax schedule to each loan at its own size, instead of at an assumed representative loan-to-value ratio, gives a gross schedule-implied borrower charge of 1.41% of price for houses and 1.28% for condominiums, before credits, exemptions and consolidation (CEMA) adjustments. Article 11 of the Tax Law reaches mortgages of real property; a co-operative apartment is not real property, and a share loan secured against it is personalty outside this tax. The calculation applies the published schedule to recorded principal. It does not measure taxable new money or apply credits and exemptions, it is not inferred from prices, and it does not estimate economic incidence. With transactions held fixed, the gross schedule-implied charge on residential purchase money in the four ACRIS boroughs between \$300,000 and \$6,000,000 comes to about \$244 million a year, a fixed-sample benchmark, and continuous schedules that raise the same static total are costed for comparison. Because notches move transactions, the replacement schedules are continuous. These are statutory calculations. A welfare analysis would also need behavioral responses to such schedules and the economic incidence of each tax.

A companion paper (Loschi 2026c) estimates a repeat-sales association between financing

and house prices in the same city and finds that it depends heavily on who pays cash. That association does not identify the recording tax’s incidence, and the statutory calculations here do not depend on it. A second companion paper (Loschi 2026b) shows that some ACRIS deeds carry consideration set by foreclosure transfer-tax rules rather than negotiation. Its classifier has been checked against AI readings of a sample of deed documents, which measures agreement with those readings rather than accuracy against a verified record; it is not used to screen the main price sample, and Section 4.10 uses it only as a robustness check.

1.2 Relation to the literature

The market-level object this paper sets aside is the one Reher and Valkanov (2024) measure nationally and the companion paper measures for New York (Loschi 2026c): a financing-associated price gap whose division among execution risk, liquidity, changing condition and selection remains open. This paper does not try to resolve that question. Neither the statutory charge, computed from a published schedule and observed loan amounts, nor the relocation of transactions at notches depends on how that gap is interpreted.

The notch evidence sits with Kopczuk and Munroe (2015), who study the New York and New Jersey transfer-tax notch and find both price bunching below the threshold and a shortfall of transactions above it that exceeds the bunching, so part of the market unravels instead of simply repricing. Best and Kleven (2018) document bunching at UK stamp-duty thresholds, Kleven and Waseem (2013) supply the notch estimator, Saez (2010) the kink design it extends, and Kleven (2016) a review of both. Best et al. (2020) use notches in UK mortgage rates at loan-to-value limits to estimate the elasticity of intertemporal substitution. Other natural experiments point the same way. Slemrod et al. (2017) find sale prices moved to the lower-rate side of a notched change in Washington, D.C.’s transfer taxes, weaker evidence of sales retimed to beat the increase, and no lock-in of volume away from the notch. Dachis et al. (2012) find that Toronto’s 1.1% land transfer tax, introduced in 2008, reduced sales of single-family homes by 15% and lowered prices by about the amount of the tax. Besley et al. (2014) study the United Kingdom’s 2008–09 stamp duty holiday. The average tax saving of about £1,500 lowered buyers’ after-tax price by about £900, so roughly sixty percent of the gain went to buyers, and most of the extra sales reversed once the holiday ended. Green et al. (2026) show that the transfer tax Los Angeles imposed in 2023 on sales above \$5,000,000 sharply reduced transactions and eroded property-tax revenue through deferred reassessment. This paper adds three things. The 2019 statutory change turns a fitted-counterfactual design into a before-and-after comparison at the same price point, with an always-treated control.

The full 2019 schedule, seven thresholds introduced on the same date, holds the tax’s age fixed, and at the thresholds with enough sales the sampling bootstrap rejects a response proportional to the buyer’s rate increment. Neither design identifies the effect of statutory incidence. The paper also counts the sales that moved below each notch and prices the tax they avoided. The contribution is the same-price comparison around a schedule introduced at one date, including its reference-price checks and conditional dollar magnitudes. It does not establish a new general theory of bunching.

The statutory calculation isolates one observable part of the transaction cost difference. Whether a change in that liability improves welfare requires economic-incidence and behavioral evidence beyond this paper.

Section 2 sets out the tax schedules. Section 3 describes the data and the record linkage. Section 4 presents the notch evidence. Section 5 applies it to the mortgage recording tax: it measures the borrower-side charge on observed loans and costs a notch-free change with transactions held fixed. Section 6 states what this paper cannot settle. Section 7 concludes.

2 The tax schedules

2.1 The thresholds

Four features of the transfer-tax schedule matter for Section 4.

§1402-a, the mansion tax, imposes 1% on residential conveyances of \$1,000,000 or more. It is a **notch**, not a bracket: the tax applies to the whole consideration, so a sale at \$1,000,000 owes \$10,000 that a sale at \$999,999 does not. The threshold has been in place, unchanged, since 1989.

§1402-b, a supplemental tax on New York City residential conveyances, is graduated across seven brackets: 0.25% for consideration of \$2,000,000 or more, 0.50% from \$3,000,000, 1.25% from \$5,000,000, 2.25% from \$10,000,000, 2.50% from \$15,000,000, 2.75% from \$20,000,000 and 2.90% from \$25,000,000. Each rate applies to the whole consideration, so every step is a notch, and the grantee pays it. It took effect for conveyances on or after **1 July 2019**, with conveyances made pursuant to binding written contracts entered into on or before 1 April 2019 excluded.

The state base rate under §1402 rises from 0.40% to 0.65% for residential conveyances of \$3,000,000 or more, enacted in the same 2019 budget and effective on the same date. The \$3,000,000 threshold therefore carries a **half-point** discontinuity, not a quarter-point: the supplemental rate rises 0.25 points and the base rate rises 0.25 points at the same consideration.

The city’s own Real Property Transfer Tax steps from 1% to 1.425% at \$500,000 for one- to three-family homes, condominiums and co-operative units. This 0.425-point notch has been in force since 1989, below the price range the 2019 reform touched. Section 4.5 reports it alongside the other three and explains why its one-dollar-below comparison measures round-number pricing rather than avoidance.

The 2019 reform is the source of identification in Section 4. It created seven discontinuities on a known date, from \$2,000,000 to \$25,000,000, and left the \$1,000,000 and \$500,000 discontinuities untouched.

2.2 The mortgage recording tax

New York imposes a tax on the recording of a mortgage on real property (New York State 2026). The components are a basic tax of 50 cents per \$100 under Tax Law §253(1); a special additional tax of 25 cents per \$100 under §253(1-a); and, within the Metropolitan Commuter Transportation District, an additional 30 cents per \$100 under §253(2). Section 253-a authorises a city of one million or more to impose its own by local law, and New York City does.

For a one- to three-family dwelling or an individual condominium unit the aggregate rate is 2.05% on mortgages below \$500,000 and 2.175% at or above. The special additional tax of 25 cents per \$100 is, by the terms of §253(1-a), payable by the mortgagee for property containing not more than six residential dwelling units. What remains with the borrower is therefore **1.80% below \$500,000 and 1.925% at or above**.

A purchaser paying cash records no mortgage and pays none of it.

The magnitude is not trivial relative to the transaction. On a \$900,000 condominium purchased with a \$720,000 mortgage, the unadjusted borrower-side schedule implies \$13,860 before credits, exemptions or consolidation adjustments. Two buyers offering the same price for the same property face different statutory closing charges, and the law assigns the difference to the one who records a purchase mortgage.

2.3 What statutory liability does and does not show

The schedule assigns a charge to recorded purchase borrowing. Two buyers paying the same price can therefore have different statutory closing charges if one records a taxable mortgage and the other does not. This comparison holds the sale price and financing terms fixed. Sellers, lenders and buyers may adjust those terms, so it does not measure the resulting economic burden, the change in transaction volume, or the welfare effect of the tax.

A companion paper estimates a financing-associated price gap in the same city (Loschi 2026c). Repeat-sales associations do not supply the counterfactual prices needed to assign the recording tax’s incidence. The statutory computations below stand independently of that price-gap estimate.

2.4 The co-operative gap

Article 11 of the Tax Law, which imposes the mortgage recording tax, reaches mortgages *of real property*. A co-operative apartment is not real property: the purchaser acquires shares in a corporation and a proprietary lease. A loan secured by those shares is a security interest in personalty, perfected by filing under the Uniform Commercial Code, and no mortgage recording tax attaches.

The consequence is that a financed co-operative purchase escapes the tax entirely while a financed condominium purchase of the same value does not. Two buyers of equivalent apartments, both borrowing, face a difference of nearly two points depending on the form of ownership.

This gap has an empirical counterpart that Section 3 documents: because share loans are not recorded as real-property mortgages, co-operative purchases appear in the data as though they were almost all cash. They are not. The same statutory boundary that exempts them from the tax also places individual share financing outside this paper’s real-property mortgage linkage. Separate personal-property filings can contain information on share loans.

By contrast, the mansion tax under §1402-a expressly defines residential real property to include “a cooperative apartment unit.” The two articles thus draw the boundary of taxable residential property differently: Article 31 reaches co-operative units, and Article 11 does not.

3 Data

Everything used here is public, free, and requires no key or licence.

3.1 Transactions

The New York City Department of Finance publishes annualized calendar sales through NYC Open Data (resource `w2pb-icbu`), covering 1 January 2016 to 31 December 2025. The residential building classes are one-, two- and three-family dwellings, tax-class-1 condominiums, walkup and elevator condominiums and co-operatives, and condominium co-operatives. With recorded consideration between \$300,000 and \$6,000,000 they give 431,471 transactions.

Section 4.6 removes the upper limit by retrieving the same resource with the same filter again: 442,497 transactions above \$300,000, whose \$300,000–\$6,000,000 slice reproduces the 431,471 exactly, sale for sale.

One formatting detail in the sales file matters. Calendar year 2016 writes the building-class category with a double space after the two-digit code (“01 ONE FAMILY DWELLINGS”); 2017 onward use a single space. A filter written on the full literal strings silently drops the whole of 2016, the earliest pre-treatment year for Section 4, and would create a before-and-after difference out of a formatting change. The filter used here matches on the two-digit code.

3.2 Financing

New York City’s Automated City Register Information System (ACRIS) records every instrument affecting real property **in Manhattan, the Bronx, Brooklyn and Queens**. Richmond County is not in it: Staten Island instruments are recorded by the Richmond County Clerk, and the city’s own guidance directs users there. Applying a deed-based method to Staten Island therefore returns a financed share of 0.0% on 41,445 sales, an artefact of the administrative boundary. Staten Island is therefore excluded from the loan-level statutory-burden measurement of Section 5.1. It is retained in Section 4, which studies prices rather than financing and needs no ACRIS match.

Two ACRIS tables are used. **Real Property Master** (bnx9-e6tj) supplies document type, consideration or loan amount, document date and recording date; restricted to plain DEED and MTGE documents recorded between 2015 and 2026 it gives **1,354,279 records**. **Real Property Legals** (8h5j-fqxa) supplies borough, block and lot for each document; restricted to the same period, **5,460,074 rows**, of which 3,609,058 resolve to a single parcel. Documents touching several parcels are dropped, since a mortgage spanning multiple lots cannot be attributed to one sale.

Restricting to plain DEED excludes DEED, LE (life estate), DEED, TS, DEEDO, DEED, RC and DEED COR. Correction deeds and life-estate transfers are therefore out of the sample before any further cleaning. This restriction does not remove foreclosure transfers: referee’s deeds and deeds in lieu of foreclosure are recorded as plain DEED and must be identified from their parties (Loschi 2026b). Section 6 returns to what this implies.

A purchase is classified **financed** if a same-parcel MTGE instrument is dated within -15 to $+90$ days of the sale, using the document date where available and the recording date otherwise, and **cash** otherwise. Cash therefore means that no such mortgage was matched, not verified absence of financing, and the match does not establish a mortgage’s purpose.

Purchase-money mortgages are normally recorded with or shortly after the deed. Two variants are reported throughout, -15 to $+30$ and -15 to $+180$. A refinancing that falls inside the window and a purchase mortgage that falls outside it both cause classification errors, which can run in either direction.

3.3 The condominium problem, and its solution

The parcel identifier does not join directly for condominiums. The Department of Finance bills a condominium unit under a billing lot in the 7501-and-above range, while ACRIS records instruments against the unit lot (1001, 1002 and so on). The two identifiers never meet, and a naive join returns a financed share of 0.0% for every condominium category. PLUTO does not bridge them: it contains 11,253 billing lots and only 2,188 unit lots, and its `condono` field numbers a condominium within a borough rather than linking its constituent units.

The route that works uses the deed itself. **The deed for any given sale is in ACRIS, carrying the consideration and the date, and its legal record carries the unit lot.** A Department of Finance sale can therefore be matched to its own deed on borough and block, consideration within 0.5%, and date within 45 days; the match reveals the unit lot, and the financing question is then asked at the correct parcel. A match is accepted only when it is unique. Where several deeds on a block share a consideration and a date, the sale is dropped rather than guessed. In a large condominium that ambiguity is common, and a wrong lot produces a wrong financing flag.

The method can be validated, because for one- to three-family dwellings the two identifiers already agree and the recovered lot can be checked against the truth. Of 211,394 house sales attempted, 160,918 find a unique deed; 44,954 find none, 3,042 have no parcel identifier and 2,480 are ambiguous. Among the unique matches the method recovers the correct lot in 160,339 cases, 99.6%.

Applied to condominiums it matches 82,488 sales: 71.4% of the 115,587 attempted, or 73.8% of the 111,743 that carry a usable parcel identifier at all. A sale with no identifier cannot be attempted at all, which is why the two percentages differ. The remainder splits into 20,004 with no deed found, 9,251 ambiguous and 3,844 with no identifier. The financed share among matched condominium sales is 58.3%, against 77.0% for houses in the same four boroughs and 0.0% from a naive join. (The house figure falls to 61.7% when Staten Island is included, for the reason given in Section 3.2; the condominium figure does not move, because no Staten Island condominium sale is in the matched set.)

3.4 The analysis sample

The deed-based panel contains **249,395 transactions in the four ACRIS boroughs**: 166,907 houses and 82,488 condominiums, on 2016–2025 sale dates, each with a price, a date, a parcel and a financing flag. A further 41,445 Staten Island house sales are matched but carry no usable financing flag and are excluded. This panel, restricted to purchases with an observed loan amount, is what Section 5.1 measures the recording-tax burden on. Section 4 works instead on the full 431,471-transaction sales file, and Section 4.6 on its extension above \$6,000,000, since both need prices and dates but no financing flag.

4 Transaction shifting at the 2019 notches

4.1 Why shape is a separate question from rate

The paper does not determine an optimal tax rate. This section studies the observed response to liability jumps at fixed price thresholds. That response is relevant to the design of a possible replacement schedule, but cannot by itself identify behavior under a continuous tax.

A notch is a discrete jump in liability at a price point, applied to the whole consideration rather than to the increment. It can create an incentive to change the recorded price or timing near the threshold. Depending on bargaining and legal constraints, some transactions may be delayed or fail to occur. Kleven and Waseem (2013) formalise the estimator, extending to notches the bunching design Saez (2010) developed for kinks, where the response is second-order because only the marginal rate changes; a notch changes the total liability and the response is first-order. Kopczuk and Munroe (2015), studying the New York and New Jersey transfer-tax notch, find that the shortfall of transactions above the threshold exceeds the bunching below it, so part of the market unravels instead of repricing.

In May 2026 New York legislators considered a buyer-paid 1% tax on New York City home purchases of \$1,000,000 or more made without a mortgage, which was reported a week later as likely to be dropped and was left out of the budget that lawmakers passed on 27 May (Bloomberg News 2026a;b;c). As proposed it would have been a notch at \$1,000,000. New York’s existing notches are relevant to that design, although they do not predict the response to a charge that depends on payment method.

4.2 A statutory experiment

The 2019 budget created the \$2,000,000 and \$3,000,000 discontinuities effective 1 July 2019 and left the \$1,000,000 mansion tax, in place since 1989, untouched. The design therefore has

two treated thresholds, \$2,000,000 (0.25 points) and \$3,000,000 (0.50 points, combining the supplemental step and the state base-rate rise), and one always-treated control, \$1,000,000, which should show the same response before and after. Its pre-period is a placebo at the same price point. That is stronger than a placebo at a different price point, because focal-price behaviour at round numbers is held fixed.

Conveyances under binding contracts signed on or before 1 April 2019 were grandfathered, so April to December 2019 is dropped as a transition window rather than assigned to either period. The pre-period runs January 2016 to March 2019 (140,538 sales) and the post-period January 2020 to December 2025 (257,855 sales). Sections 4.3–4.5 use the two thresholds below \$6,000,000; Section 4.6 extends the design to the reform’s five higher ones.

4.3 Bunching, differenced

The estimator follows Kleven and Waseem (2013) and Kopczuk and Munroe (2015): a fifth-degree polynomial counterfactual fitted in a $\pm\$400,000$ window in \$10,000 bins, with an asymmetric exclusion band of $-\$50,000$ to $+\$100,000$ because the hole above a notch is wider than the pile below it, and indicators for exact multiples of \$100,000, \$50,000 and \$25,000 so that the counterfactual absorbs focal mass rather than attributing it to the tax. The bin containing the threshold is assigned to the taxed side, since the notches apply at “or more”.

4.4 The clearest evidence: focal prices

Round numbers attract transactions. A notch at a round number should destroy that attraction, because the round number is the first taxed price, and should displace the mass one dollar below.

4.5 A fourth notch, and what the four together reveal

New York City’s own Real Property Transfer Tax charges 1% on consideration of \$500,000 or less and 1.425% above it, for one- to three-family homes, condominiums and co-operative units. This 0.425-point notch sits below the mansion tax, where most buyers transact. Because the higher rate starts above \$500,000 rather than at it, a sale at exactly \$500,000 already pays the lower rate, and the one-dollar comparison used at the other thresholds does not measure avoidance there. The row is reported for completeness and is not used in any inference.

Pricing one dollar below a threshold requires someone in the transaction to notice it and

Table 1: Bunching at the \$1,000,000, \$2,000,000 and \$3,000,000 thresholds, differenced across the 2019 reform

<i>Panel A. Share of counterfactual mass missing above the threshold</i>					
Threshold	Status in 2019	Pre	Post	Difference	95% CI
\$1,000,000	taxed throughout	0.508	0.513	+0.005	[−0.020, +0.034]
\$2,000,000	untaxed → taxed	0.094	0.314	+0.220	[+0.133, +0.320]
\$3,000,000	untaxed → taxed	−0.117	0.344	+0.461	[+0.287, +0.645]
\$1,500,000	never taxed	−0.049	−0.076	−0.027	[−0.103, +0.055]
\$1,750,000	never taxed	−0.004	+0.041	+0.045	[−0.028, +0.136]
\$2,250,000	never taxed	−0.018	−0.094	−0.076	[−0.206, +0.053]
\$2,500,000	never taxed	−0.080	−0.096	−0.016	[−0.148, +0.148]
<i>Panel B. The same result with no counterfactual fitted</i>					
Threshold	R pre	R post	Log change	95% CI	
\$1,000,000	0.289	0.302	+0.042	[−0.011, +0.092]	
\$2,000,000	0.738	0.531	−0.328	[−0.431, −0.222]	
\$3,000,000	0.933	0.517	−0.591	[−0.764, −0.425]	
\$1,500,000	0.763	0.792	+0.036	[−0.031, +0.101]	
\$1,750,000	0.777	0.771	−0.008	[−0.093, +0.077]	
\$2,250,000	0.880	0.916	+0.040	[−0.071, +0.157]	
\$2,500,000	0.882	0.900	+0.021	[−0.097, +0.148]	

Notes. Panel B defines $R(X)$ as the number of transactions in $[X, X + 100,000)$ divided by the number in $[X - 150,000, X - 50,000)$. A notch at X predicts lower R , but changing density or transaction composition can also move it. The \$1,000,000 row checks the design. Nothing changed at that threshold in 2019, and the estimate stays at 0.508 before and 0.513 after, on 64,066 and 134,996 transactions in the estimation window. The small fixed-window control change reduces, without eliminating, the concern that the before-and-after comparison picks up general drift in the price distribution. Same conclusion in both panels, and Panel B needs no polynomial. This table uses a fixed $\pm\$400,000$ window. From Table 4 on, windows scale with the threshold ($\pm 0.2X$) so that one design covers every threshold. The two are identical at \$2,000,000; at \$3,000,000 the scaled window is wider and the change is +0.321, the figure used from then on.

to insist. The well-populated buyer-paid notches in this table show transaction shifting. The table does not compare an otherwise equivalent seller-paid notch or predict the response to a cash-only charge.

4.6 The full 2019 schedule

Four thresholds are a small basis for any claim about what drives the response, and the four above differ in more than who pays: the \$1,000,000 notch has been in force for three decades,

Table 2: Focal prices, before and after the 2019 reform

Price point	Spike at exactly X , relative to local density		Sales at $X - 1$	
	Before Apr 2019	From Jan 2020	Before	From Jan 2020
\$1,000,000 — taxed throughout	0.1	0.1	156	543
\$2,000,000 — treated	7.8	1.6	4	131
\$3,000,000 — treated	7.0	1.5	2	68
\$1,500,000 — never taxed	5.2	6.2	3	4
\$2,500,000 — never taxed	7.8	10.6	1	0

Notes. The two periods are **before 1 April 2019** and **from 1 January 2020**, leaving out the nine months in between: conveyances under binding contracts signed on or before 1 April 2019 were grandfathered, so the intervening sales are a mixture of treated and untreated and belong to neither period. The counts therefore do not sum to the sample totals used in Table 3, which needs no before-and-after split. This requires no estimator, no counterfactual and no distributional assumption. It is a count of transactions at five prices.

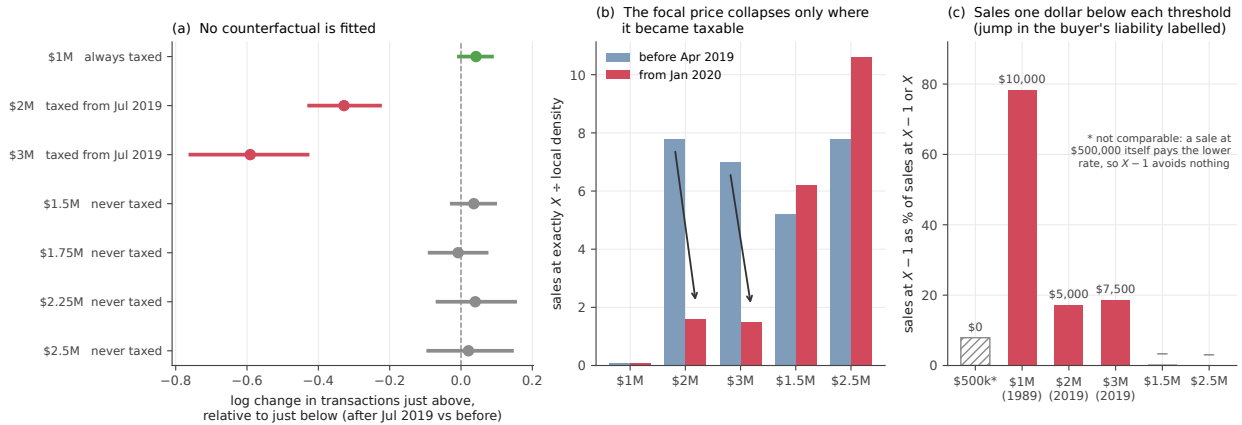


Figure 1: The §1402-b notches switched on 1 July 2019. Three descriptive measures show a response at the two newly taxed thresholds shown, and not at the one that did not change. Panel (a), post-minus-pre log above/below ratios, with no fitted counterfactual (Table 1, Panel B). Panel (b), the focal-price spike at exactly X relative to local density (Table 2). Panel (c), sales one dollar below the threshold as a share of sales at $X - 1$ or X , labelled with the jump in the buyer's own liability (Table 3); the \$500,000 bar is not comparable, because a sale at exactly \$500,000 already pays the lower rate, and the thresholds differ in age as well as in the size of the jump.

Table 3: Four residential notches, and the buyer’s share of each

<i>Panel A. Charm pricing at each threshold, all transactions 2016–2025</i>					
Threshold	Notch	In force since	Sales at $X - 1$	Sales at X	Charm share
\$500,000 (city RPTT) [†]	0.425 pt	1989	249	2,911	7.9%
\$1,000,000 (mansion)	1.00 pt	1989	728	200	78.4%
\$2,000,000 (§1402-b)	0.25 pt	Jul 2019	140	674	17.2%
\$3,000,000 (§1402-b + base)	0.50 pt	Jul 2019	73	322	18.5%
\$1,500,000 placebo	—	—	7	1,885	0.4%
\$2,500,000 placebo	—	—	1	802	0.1%
<i>Panel B. Who owes the jump</i>					
Threshold	Total jump		Of which the buyer owes		Charm share
\$500,000 [†]	\$2,125		\$0		7.9%
\$2,000,000	\$5,000		\$5,000		17.2%
\$3,000,000	\$15,000		\$7,500		18.5%
\$1,000,000	\$10,000		\$10,000		78.4%

Notes. Charm share is sales at $X - 1$ as a percentage of sales at $X - 1$ or X . [†]**Not comparable with the other rows.** The city’s higher rate applies to consideration *above* \$500,000, so both \$499,999 and \$500,000 pay 1%, and pricing one dollar below saves nothing; the jump is avoided by a price of \$500,000 or less, which is itself a round number. The row’s charm share therefore measures round-number pricing below a salient price, not avoidance, and a boundary-aligned test would need round-number controls this table does not have. Panel B’s incidence follows the statutes: the city’s Real Property Transfer Tax and the state base tax under §1402 are imposed on the *grantor* (Administrative Code §11-2104; Tax Law §1404(a)), while the mansion tax under §1402-a and the supplemental tax under §1402-b are paid by the *grantee*; §1402-b says that “the tax imposed by this section shall be paid by the grantee.” Among the three thresholds where a price one dollar below does avoid the jump, the charm share is highest at the oldest, \$1,000,000. Section 4.6 tests on the 2019 schedule whether the size of the buyer’s jump matters.

the 2019 notches for six years. The 2019 reform supplies a cleaner test. Its supplemental tax steps up at seven prices on the same date, all paid by the buyer, with rate increments from 0.15 to 1.00 points and dollar jumps from \$5,000 to \$100,000. Holding the tax’s age fixed, a response that tracked the buyer’s liability should grow across these thresholds.

It does not grow in proportion. At \$5,000,000, where the buyer’s rate rises by 0.75 points, transactions relocate on every measure in Table 4: bunching rises by 0.219, the ratio of sales just above to just below the threshold falls by 0.466 in logs, and sales at \$4,999,999 go from none to 26. The point estimate matches the one at \$2,000,000, where the buyer’s rate rises by 0.25 points and the jump is \$5,000 rather than \$37,500: bootstrapping the two thresholds jointly, the difference is -0.001 [-0.190 , $+0.219$]. A response proportional to the buyer’s rate would make the \$5,000,000 change three times the \$2,000,000 one; the shortfall from that benchmark is -0.441 [-0.769 , -0.118], which rejects proportionality ($p = 0.008$). Adding

Table 4: The full 2019 schedule: every threshold, before and after the reform

Threshold	Buyer's jump (rate / dollars)	Sales in window pre/post	Bunching, post – pre [95% CI]	$\Delta \log R$, no counterfactual [95% CI]	Sales at $X - 1$ pre $\rightarrow$ post
<i>Panel A. Thresholds created by the 2019 reform</i>					
\$2,000,000	0.25 pt / \$5,000	9,299/23,022	+0.220 [+0.136, +0.316]	–0.328 [–0.434, –0.223]	4 $\rightarrow$ 131
\$3,000,000	0.25 pt / \$7,500 ^a	5,268/12,374	+0.321 [+0.183, +0.469]	–0.348 [–0.483, –0.214]	2 $\rightarrow$ 68
\$5,000,000	0.75 pt / \$37,500	2,392/5,791	+0.219 [+0.046, +0.402]	–0.466 [–0.663, –0.270]	0 $\rightarrow$ 26
\$10,000,000 ^b	1.00 pt / \$100,000	817/1,892	+0.052 [–0.297, +0.399]	–0.258 [–0.622, +0.106]	3 $\rightarrow$ 7
\$15,000,000 ^b	0.25 pt / \$37,500	415/1,207	–0.838 [–1.539, –0.238]	+0.198 [–0.235, +0.631]	0 $\rightarrow$ 2
\$20,000,000 ^b	0.25 pt / \$50,000	221/1,186	+0.911 [+0.517, +1.713]	–2.198 [–2.812, –1.584]	0 $\rightarrow$ 0
\$25,000,000 ^b	0.15 pt / \$37,500	172/647	–0.217 [–0.772, +0.592]	–1.124 [–1.895, –0.353]	0 $\rightarrow$ 0
<i>Panel B. Round-number placebos, never taxed</i>					
\$4,000,000	none	3,348/7,753	–0.074 [–0.225, +0.075]	+0.036 [–0.126, +0.198]	1 $\rightarrow$ 5
\$6,000,000	none	1,909/4,540	–0.068 [–0.324, +0.213]	–0.047 [–0.270, +0.176]	0 $\rightarrow$ 1
\$7,000,000	none	1,509/3,550	–0.270 [–0.588, +0.040]	+0.010 [–0.224, +0.245]	0 $\rightarrow$ 0
\$8,000,000	none	1,234/2,849	–0.132 [–0.393, +0.144]	+0.275 [+0.007, +0.544]	0 $\rightarrow$ 0
\$12,000,000 ^b	none	581/1,360	+0.388 [–0.222, +1.211]	–0.294 [–0.693, +0.106]	0 $\rightarrow$ 0

Notes. Pre is January 2016 to March 2019; post is January 2020 to December 2025. Windows scale with the threshold so one design applies across the schedule, and at \$2,000,000 every window equals the one used in Table 1. Bunching is the Kleven–Waseem share of counterfactual mass missing above X (window $\pm 0.2X$, bins of 0.005 X rounded down to \$10,000, \$25,000, \$50,000 or \$100,000, exclusion band $(-0.025X, +0.05X)$, degree-five polynomial with round-number indicators), post minus pre, with bootstrap percentile intervals. R is sales in $[X, 1.05X)$ over sales in $[0.925X, 0.975X)$; its log change has a Poisson interval. ^aAt \$3,000,000 the seller's base rate also rises 0.25 points; the wider scaled window gives a smaller change than Table 1's fixed one (+0.321 against +0.461). ^bFewer than 1,000 pre-reform sales in the estimation window. On samples this small the estimator returns large, individually significant values of both signs at treated thresholds (\$15,000,000 and \$20,000,000), so no conclusion is drawn from these rows. The price sample extends above the \$6,000,000 cap of Table 1; its \$300,000–\$6,000,000 slice reproduces that table's 431,471 sales exactly.

the seller-paid 0.25 points at \$3,000,000 raises the response by +0.101 [–0.064, +0.278], not significantly, and equal responses at all three thresholds are not rejected ($p = 0.53$). Failing to reject equality is not evidence of equality: the interval on the \$5,000,000 minus \$2,000,000 difference admits differences almost as large as the \$2,000,000 response itself. The bootstrap also resamples sales within threshold and period, so repeat sales of one property in different windows are treated as independent. Section 4.9 repeats these tests against placebo noise at matched sample sizes. The estimates keep their direction there, but the evidence against the proportional benchmark is weaker and depends on the reference set. Every never-taxed placebo with enough sales has an interval that includes zero on the bunching measure. Above \$5,000,000 the bunching estimator becomes unreliable: fewer than 1,000 sales fall in each pre-reform window, and it returns large values of both signs. The paper calls a threshold with fewer than 1,000 pre-reform sales in its estimation window thin, and the others well-populated. The simplest count still reads. Comparing sales in the one per cent of price

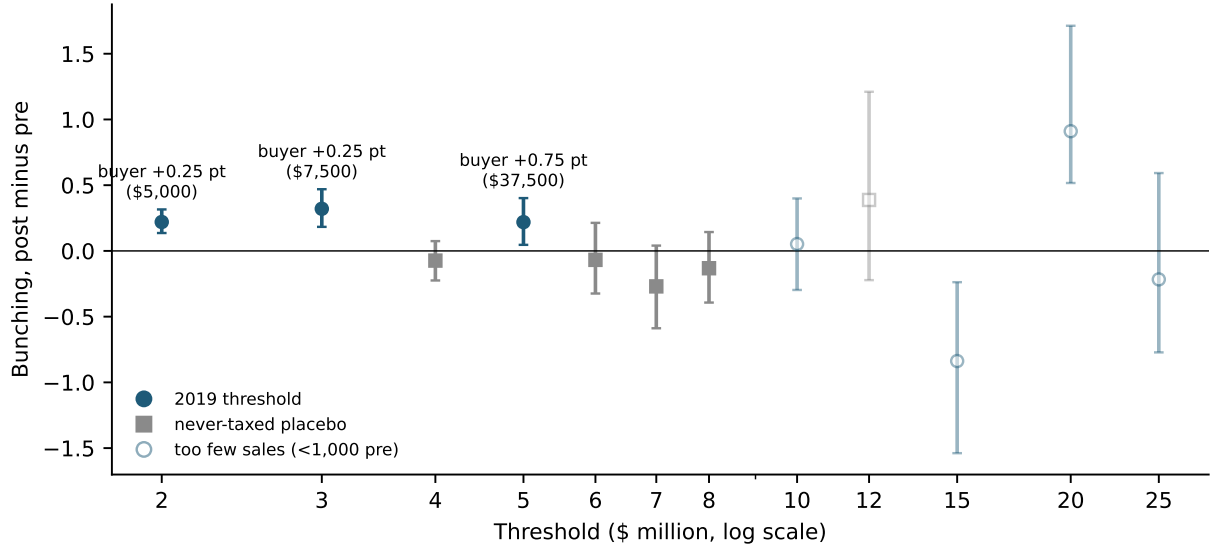


Figure 2: The response to a buyer-paid notch does not rise in proportion to the buyer’s jump. Change in bunching after the 2019 reform at each threshold of the schedule, with 95% bootstrap intervals (Table 4). Filled circles are reformed thresholds with enough sales to estimate; squares are never-taxed placebos; hollow markers have fewer than 1,000 pre-reform sales in the window.

below each threshold with the one per cent above, the post-reform shift is +0.93 log units at \$10,000,000 and +1.31 at \$15,000,000, both with intervals excluding zero, against +1.68 at \$2,000,000; every placebo is null (Supplemental Appendix Table A3). The larger jumps again have no larger point estimates. Even so, the \$10,000,000 threshold, whose one-point jump matches the mansion tax’s, is too thin to separate a notch’s tenure from its size.

What survives is narrower than the four-threshold reading. Every buyer-paid notch with enough data relocates transactions, and between 0.25 and 0.75 points the response does not rise in proportion to the rate: the proportional benchmark is rejected, though equality is not established. The design does not show that incidence matters. The only seller-paid comparisons are the \$500,000 threshold, where the one-dollar comparison measures no avoidance, and the seller-paid addition at \$3,000,000, whose effect is estimated imprecisely. For design, the useful result is the first: within the observed range a lower buyer-paid rate did not produce a proportionally smaller response (the sampling bootstrap rejects proportionality; the placebo comparisons of Section 4.9 are weaker), and even a quarter-point notch relocates transactions.

4.7 Who responds

Section 3.2 divides buyers by payment method. Here that division asks who avoids a notch; it says nothing about any price gap between the two groups.

Splitting the charm-price counts of Table 3 by financing status, financed buyers price one dollar below the threshold more often than cash buyers at all four thresholds, by factors of 1.2 to 2.9, while both never-taxed placebos show nothing for either type. Supplemental Appendix Table A2 reports the counts. Three things keep this from being a result. At \$500,000 the charm price avoids no tax (Table 3), so there the split measures charm pricing, not avoidance. The cells are small: the \$3,000,000 comparison rests on 12 of 110 against 40 of 125, so the ordering is better read as a pattern than as an estimate. And the corresponding split of the *bunching* magnitudes is a null: applying the Kleven–Waseem estimator separately by buyer type gives differences of +0.063, +0.053, +0.165 and -0.069 at the four thresholds, but the never-taxed placebos return +0.054, +0.054, +0.004 and -0.137 by the same procedure. That split is not distinguishable from specification noise and should not be read as evidence. The charm-price comparison survives only because it is a count of transactions at two prices with no counterfactual to misspecify.

The pattern fits a notch that changes financed buyers’ behaviour more, so that the cost of relocating transactions falls mostly on the market’s financed side, but it does not establish it. Pricing below a notch is *successful avoidance*, so on this evidence cash buyers pay the notch more often; the claim is about responsiveness, not burden. Two readings of *why* are available. Payment method may proxy liquidity: buyer-side transfer taxes are paid in cash at closing and cannot be financed, and a financed buyer has already committed cash to the down payment. Or a financed transaction involves more professionals (a lender, an appraiser, an attorney preparing a cash-to-close statement), who are more likely to notice the threshold. Loan-to-value separates the two, because the liquidity reading predicts that the most leveraged buyers avoid the notch most. Among 627 financed house purchases recorded at exactly X or $X - 1$ at the three buyer-paid thresholds, with threshold and year fixed effects, the probability of the charm price rises by 0.3 percentage points per ten points of loan-to-value (standard error 0.9), and buyers borrowing 80% or more of the price choose it 12 points *less* often (standard error 3.4). **The loan-to-value pattern does not support the specified liquidity prediction; professional intermediation remains one possible explanation.** Adding the 210 financed condominium purchases at these prices, with loans from the linked dataset (Loschi 2026a), changes nothing. Across both types the charm price rises by 0.2 points per ten points of loan-to-value (standard error 0.8), and buyers borrowing 80% or more choose it 13 points less often (standard error 3.0; Supplemental Appendix Table A9).

4.8 Robustness, and two concessions

The differenced result is stable across period definitions. The baseline, dropping COVID, a clean 2016–2018 pre-period, a tight 2017–18 against 2020–21 window and a late 2023–2025 post-period all leave the \$2,000,000 estimate between +0.202 and +0.221 and the \$3,000,000 estimate between +0.311 and +0.531, while the always-treated control never moves beyond ± 0.014 . Supplemental Appendix Table A1 reports them.

Nominal prices, and bracket creep. All estimates use nominal consideration, because statutory liability under §§1402-a and 1402-b attaches to nominal consideration; deflating would smear each threshold across a moving band of real values and destroy the focal-point behaviour that identifies the response. Price appreciation over 2016–2025 does shift the density rightward. The share of transactions above \$2,000,000 rises from 8.4% to 12.3%, and above \$1,000,000 from 25.5% to 38.5%, which could affect local price density. The estimator is a local ratio, so a common rightward shift enters numerator and denominator alike; the always-treated \$1,000,000 threshold experienced proportionally the same creep and returns +0.005; and the tight 2017–18 against 2020–21 comparison, over which cumulative appreciation is small, leaves the \$2,000,000 response at +0.202. These checks reduce, but do not eliminate, concern that changing transaction composition contributes to the before-and-after pattern.

Two concessions the larger sample forces. The level estimates overstate. In the pre-period, when no notch existed at \$2,000,000, the estimator returns a missing-mass share of 0.094 with a bootstrap interval excluding zero. It finds a hole where there is no tax. The credible figure is the difference, 0.220, not the level of 0.314. **And the bootstrap does not cover specification error.** Three of the four post-period placebos return nominally significant bunching or anti-bunching. The bootstrap covers sampling uncertainty conditional on the estimator, not specification error. Section 4.9 therefore adds descriptive placebo comparisons.

4.9 Placebo-calibrated inference and timing

Two questions remain: whether the uncertainty is priced correctly, and whether the change happened in 2019 rather than drifting in over the decade.

Inference against never-taxed prices. Table 5 judges each estimate against the same statistic computed where no tax exists. The placebo set is every \$50,000 multiple between

\$1,200,000 and \$6,000,000 whose estimation window ($\pm 0.2X$) contains no threshold of the city or state schedules: sixteen prices, from \$1,300,000 to \$1,650,000 and from \$3,800,000 to \$4,150,000. Their post-minus-pre bunching changes average -0.009 , with a standard deviation of 0.047 and a maximum of $+0.080$. The changes at \$2,000,000, \$3,000,000 and \$5,000,000 are each larger than all sixteen, and each log change in R is lower than all sixteen. Neighbouring windows share most of their sales, so the sixteen describe two stretches of the price distribution rather than sixteen independent draws. The table therefore reports each ranking as $(1+k)/17$, where k counts placebos at least as extreme; with none that is $1/17 = 0.06$. A calibrated rank test would require a justified exchangeability or randomization argument for the treated and placebo statistics. This design does not establish one, and the overlap between windows makes it doubtful, so we read the rank as a reference comparison.

Noise depends on how many sales a window holds, and the placebo windows differ in size. To compare like with like, every placebo window with more sales than a treated threshold's is resampled with replacement, fifty times, to exactly that threshold's pre- and post-reform counts. The pooled draws mix sampling error at the treated threshold's own sample size with the specification error of the placebo locations. Against them the \$2,000,000 and \$3,000,000 changes lie 3.7 and 4.3 standard deviations from the draws' mean and exceed every draw. The \$5,000,000 change, estimated on about a quarter of the \$2,000,000 sample, lies 1.9 standard deviations out, and 4% of the draws are as large. These tail shares describe a mixture over few locations: all eight placebos large enough to match \$2,000,000 and \$3,000,000 lie between \$1,300,000 and \$1,650,000. They are reference comparisons, not calibrated p -values. The tests of Section 4.6 keep their direction. The sampling bootstrap rejects proportionality ($p = 0.008$). About 4.8% of the selected placebo contrasts are at least as extreme (a share of 0.048; 5.3% using only placebo pairs whose windows do not overlap), so these comparisons provide weaker, reference-dependent evidence. The \$5,000,000 minus \$2,000,000 difference remains uninformative (0.99), and adding the seller-paid step at \$3,000,000 does not change the response detectably (0.31).

Three qualifications apply. The matched tests rest on few placebos, eight for \$2,000,000 and \$3,000,000 and sixteen for \$5,000,000, so draws from one placebo share its specification error, and neighbouring placebos share sales. Version 2.4 drew these subsamples without replacement. Each draw then took up to three-quarters of a placebo window, so the draws varied less than repeated samples of that size from the placebo's price distribution would, by about a quarter at \$2,000,000; drawing with replacement approximates such repeated samples. It does not make the placebos exchangeable with the treated windows or remove the dependence created by overlapping windows and repeat sales. Adding the 41 thinner-market placebos between \$6,300,000 and \$8,300,000 widens the placebo standard deviation

to 0.159: against that wider set the bunching changes have reference ranks of 0.07, 0.03 and 0.07, the log changes in R are still lower than every placebo (rank 0.017), and the proportional contrast is less unusual (a share of 0.33). And the fixed-window estimator of Table 1, applied at the 44 eligible \$50,000 multiples from \$1,450,000 to \$6,000,000, gives reference ranks of 0.07 at \$2,000,000 and 0.02 at \$3,000,000 (0.04 and 0.02 for R); its fixed $\pm\$400,000$ window holds few sales at the higher placebo prices, which widens their spread. The always-treated control does not stand out on bunching under either estimator, but its log change in R in the scaled window, -0.095 , is lower than every placebo. The event study shows where that comes from.

Timing. Figure 3 computes both statistics year by year at each threshold. Each series is normalised to its own 2016–2018 mean, so a price point’s fixed focal-price behaviour drops out, and the grey band is the 90% range of the same series at size-matched placebos, resampled with replacement. 2019, when grandfathered contracts were still closing, is shown but not used. Before the reform the treated series stay inside the band in every year. From 2020 the change at \$2,000,000 is immediate and does not fade: $\log R$ lies below the band in all six post-reform years, between -0.27 and -0.35 , and bunching lies above it in five. At \$3,000,000 $\log R$ is outside the band in five of six years and bunching in four. At \$5,000,000, the smallest sample, $\log R$ is outside in four and bunching in one. Moving the reform back to January 2018, a date on which nothing happened, gives bunching changes of -0.112 , -0.045 and -0.022 at the three thresholds, all inside their placebo distributions (shares of 0.53 or more).

The always-treated control is flat on bunching, apart from two excursions of less than 0.03 beyond the band, in 2022 and 2024. Its $\log R$, however, falls in 2020–2022, by 0.33 in 2020, and is back near its baseline from 2023 (just above the band that year, inside it in 2024 and 2025). The dip coincides with the pandemic years and shows that the ratio can move without a change in the tax. It does not account for the treated responses: from 2023 to 2025, with the control back at its baseline, $\log R$ stays between -0.27 and -0.35 at \$2,000,000, between -0.29 and -0.55 at \$3,000,000 and between -0.31 and -0.97 at \$5,000,000. The later-year estimates at \$3,000,000 and \$5,000,000 are larger than the early ones, which fits a response that grows as a notch becomes familiar, but yearly estimates on these samples are too noisy to establish a trend.

Table 5: Placebo-calibrated inference for the 2019 notches

Threshold	Change,	Placebos at	Rank	Size-matched placebos		
	post – pre	least as extreme	$(1 + k)/17$	Resampling SD	z	As large
<i>Panel A. Bunching (share of counterfactual mass missing above X)</i>						
\$1,000,000 (control)	+0.055	2 of 16	0.18	—	—	—
\$2,000,000	+0.220	0 of 16	0.06	0.062	3.7	0 of 400
\$3,000,000	+0.321	0 of 16	0.06	0.078	4.3	0 of 400
\$5,000,000	+0.219	0 of 16	0.06	0.121	1.9	31 of 800
<i>Panel B. Log change in R, no counterfactual fitted</i>						
\$1,000,000 (control)	–0.095	0 of 16	0.06	—	—	—
\$2,000,000	–0.328	0 of 16	0.06	—	—	—
\$3,000,000	–0.348	0 of 16	0.06	—	—	—
\$5,000,000	–0.466	0 of 16	0.06	—	—	—
<i>Panel C. Tests across thresholds (bunching)</i>						
	Estimate	Bootstrap p		Size-matched placebo share		
\$5M – \$2M	–0.001	0.99		0.99		
\$5M – $3 \times$ \$2M (proportional)	–0.441	0.008		0.048		
\$3M – \$2M	+0.101	0.28		0.31		

Notes. Estimator and windows as in Table 4. Placebos are the sixteen \$50,000 multiples between \$1,200,000 and \$6,000,000 whose window $[0.8X, 1.2X)$ contains none of the thresholds at \$500,000, \$1,000,000, \$2,000,000, \$3,000,000, \$5,000,000 and above. The rank column is one-sided in the direction a notch predicts (bunching up, R down): $(1 + k)/17$, where k is the number of placebos at least as extreme. It is a reference ranking rather than an exact p -value: a calibrated rank test would need an exchangeability argument the design does not supply, and neighbouring placebo windows overlap. Size-matched placebos resample each placebo window with more sales than the treated window, with replacement, to the treated window’s pre- and post-reform counts, fifty times per placebo (eight placebos for \$2,000,000 and \$3,000,000, sixteen for \$5,000,000); z is the estimate’s distance from the draws’ mean in their standard deviations. No placebo has as many sales as the \$1,000,000 window, so the control has no size-matched row. Panel C reports the two-sided share of differences between draws at two distinct placebos that are at least as large (for the proportional contrast, 0.053 when the two placebo windows must not overlap); these shares are reference comparisons, not calibrated p -values. Bootstrap p -values are from Section 4.6. Supplemental Appendix Table A6 lists every placebo. Replication package v1.9, `placebo_inference.py`.

4.10 Where the response comes from, how large it is, and what it costs

The linked dataset (Loschi 2026a) attaches property type, borough, the seller on the deed and the deed’s classification to the same sales. Table 6 splits the response by each, against placebos recomputed within the same sample.



Figure 3: The response starts with the reform. Bunching (top) and $\log R$ (bottom) by year, each normalised to its 2016–2018 mean, at the always-treated \$1,000,000 threshold and the three thresholds created in 2019. Grey bands are the 90% range of the same statistic at never-taxed placebos resampled with replacement to each threshold’s yearly sales (for the control, the seven placebos with at least half its sales, without subsampling). Hollow points mark 2019, the transition year, which no test uses. Replication package v1.9, `event_study.py`.

By market. The response is not confined to one kind of property. Among houses the changes at \$2,000,000, \$3,000,000 and \$5,000,000 are +0.281, +0.487 and +0.502, larger than in the pooled sample, and among condominiums +0.206, +0.339 and +0.183. In Manhattan they are +0.223, +0.324 and +0.165, and in the other four boroughs +0.206, +0.370 and +0.602, the last on only 342 pre-reform sales. Of these twelve estimates only one, houses at \$2,000,000, is matched by any placebo in its own sample, by two of the sixteen. Co-operatives are the exception: their changes, +0.174, +0.161 and +0.086, are each matched by two placebos in a noisy sample.

The split also locates the control’s movement. Among houses the \$1,000,000 threshold does not move (+0.008, with eight placebos as large, and a log change in R of +0.056). The shift found in the pooled data comes from condominiums (+0.133) and from Manhattan (+0.095). Houses therefore give the cleanest comparison: their control is flat and their responses at the new thresholds are the largest. Among condominiums the new thresholds’ changes exceed their own control’s by only +0.07, +0.21 and +0.05, so the condominium evidence rests mainly on \$3,000,000.

Sales with and without a company seller. Developers set condominium prices in advance, and some sponsor sales that closed after the reform were contracted before the April 2019 cut-off. The linked deed shows whether a company sold, a proxy that includes sponsor sales and other company sales; sales without a linked deed are in neither group. In no-company-seller sales the changes at the three new thresholds are +0.349, +0.384 and +0.511, each larger than every placebo in that sample, while the \$1,000,000 control does not move (+0.022). In company-seller sales (sponsor proxy) the response is clear at \$3,000,000 (+0.431), and among condominium company-seller sales from 2022 at \$5,000,000 as well; at \$2,000,000 it stays within placebo noise in every company-seller sample (Supplemental Appendix Table A8). Company sellers' share of condominium sales near \$2,000,000 rose from 34% to 40% after the reform. Each group's estimate is normalised within its own sample, so these patterns show that shifting remains evident without company sellers; they do not divide the aggregate shift between the groups.

Foreclosure and lender deeds. Referee's deeds and deeds in lieu are recorded as plain deeds and stay in the sales file (Section 3.2). They are rare near these prices. In the estimation windows, 0.06–0.6% of sales have a deed whose amount can follow a legal rule, and 0.2–1.2% have any lender or referee party. Dropping either group leaves the changes at +0.224, +0.325 and +0.219, or +0.225, +0.324 and +0.222. The 31–42% of window sales with no linked deed, mostly co-operatives, cannot be screened this way.

How large the response is. Dividing the excess mass below a threshold and the missing mass above it by the counterfactual density of sales at the threshold expresses each as a width of price. After the reform the pile below \$2,000,000 grew by the equivalent of \$16,500 of price and the hole above by \$10,700, against a buyer's jump of \$5,000. At \$3,000,000 the figures are \$16,700 and \$21,900 against \$7,500, and at \$5,000,000 \$32,500 and \$43,700 against \$37,500. As a share of price these widths lie above every placebo, except that one placebo matches the missing mass at \$2,000,000. At \$2,000,000 and \$3,000,000 the emptied range is two to three times the range a buyer could never prefer. Sales are therefore leaving prices at which the buyer's tax saving alone would not pay for the move. That fits price renegotiation, or value moved off the deed into personal property or seller credits, better than the rounding down of sales already within a few thousand dollars of the threshold. The data cannot separate renegotiation from relabelling. At \$5,000,000 the emptied range roughly matches the jump.

The \$1,000,000 notch that Kopczuk and Munroe (2015) study gives a benchmark from the same market. In these data its pile below is equivalent to \$26,500 of price and its hole

above to \$22,800 before the reform, and \$30,000 and \$25,300 after, against a \$10,000 jump. Six years in, the 2019 notches at \$2,000,000 and \$3,000,000 have reached a similar multiple of their jump. Kopczuk and Munroe find that the missing transactions above the mansion-tax threshold exceed those bunching below it. Here the hole is larger than the pile at \$3,000,000 and \$5,000,000 but not at \$2,000,000 or \$1,000,000, and at no threshold is the difference distinguishable from zero (Table 7), so these data cannot show whether sales disappear at the new notches or only move.

What the notches cost. A sale that moves below a notch avoids at least the jump in tax. Table 7 counts the extra sales just below each new threshold: the post-reform pile less the pre-reform one, rescaled to post-reform density. About 130 extra sales a year sit just below \$2,000,000, 54 below \$3,000,000 and 20 below \$5,000,000; as a share of its window, each pile is larger than at any placebo. The data do not show each sale’s price without the reform, and extra sales below a threshold need not all have crossed it. If the estimated extra sales below each threshold relocated from above it, they avoid \$2.2 million a year at the jump alone (95% interval \$1.6–2.8 million, which covers sampling uncertainty conditional on the estimator and that assumption). In a 0.05X price-cut scenario, in which each also cut its price by the whole emptied range, the other transfer taxes on those reductions would bring the total to \$3.1 million. The reform’s schedule implies \$92.7 million a year on sales between \$2,000,000 and \$6,000,000. Under a cross-threshold relocation assumption, estimated avoidance is therefore 2–3% of schedule-implied revenue in that range.

5 Application: the mortgage recording tax

New York City’s mortgage recording tax charges a purchase that records a mortgage and nothing when it does not. This section computes the gross charge the schedule implies on observed loans and costs replacing it with a continuous purchase schedule, holding transactions fixed.

5.1 The gross schedule-implied charge on observed loans

The recording tax is levied on the loan, not on the sale, so the gross schedule-implied charge as a share of price depends on the recorded loan-to-price ratio. The ACRIIS match records the loan amount alongside the price for most financed purchases, so that ratio can be measured instead of assumed. It is measured separately for each market, since houses and condominiums have different price levels and different loan-to-value distributions and the

Table 6: Where the response comes from, and how large it is

<i>Panel A. Change in bunching, post – pre (placebos at least as large, of 16)</i>					
	\$1,000,000 (control)	\$2,000,000	\$3,000,000	\$5,000,000	Placebo SD
All sales	+0.055 (2)	+0.220 (0)	+0.321 (0)	+0.219 (0)	0.047
Houses	+0.008 (8)	+0.281 (2)	+0.487 (0)	+0.502 (0)	0.152
Condominiums	+0.133 (1)	+0.206 (0)	+0.339 (0)	+0.183 (0)	0.083
Co-operatives	+0.027 (4)	+0.174 (2)	+0.161 (2)	+0.086 (2)	0.211
Manhattan	+0.095 (0)	+0.223 (0)	+0.324 (0)	+0.165 (0)	0.061
Other four boroughs	+0.033 (3)	+0.206 (0)	+0.370 (0)	+0.602 (0)	0.138
No company seller	+0.022 (6)	+0.349 (0)	+0.384 (0)	+0.511 (0)	0.067
Company seller (sponsor proxy)	+0.090 (6)	+0.067 (6)	+0.431 (0)	+0.080 (6)	0.181
Without legal-rule deeds	+0.050 (2)	+0.224 (0)	+0.325 (0)	+0.219 (0)	0.048
Without any lender or referee deed	+0.050 (2)	+0.225 (0)	+0.324 (0)	+0.222 (0)	0.048
<i>Panel B. Size of the change, in dollars of price at the counterfactual density</i>					
	Buyer’s jump	Excess mass below X	Missing mass above X	Missing mass / buyer’s jump	
\$1,000,000	\$10,000	\$3,400	\$2,500	—	
\$2,000,000	\$5,000	\$16,500	\$10,700	2.1	
\$3,000,000	\$7,500	\$16,700	\$21,900	2.9	
\$5,000,000	\$37,500	\$32,500	\$43,700	1.2	

Notes. Estimator, windows and periods as in Table 4. Property types follow the Department of Finance building-class category (houses 01–03; condominiums 04, 12, 13, 15; co-operatives 09, 10, 17) and come from the linked dataset (Loschi 2026a), whose sales are the same records as this paper’s sample. Each row’s placebos are the sixteen of Table 5, recomputed within that row’s sample. The last two market rows split sales by whether the linked deed names a company as seller, a proxy that includes sponsor (developer) sales and other company sales; sales without a linked deed are in neither row. Deed screens use the linked dataset’s deed classifier (Loschi 2026b); sales with no linked deed (co-operatives, Staten Island sales, and houses and condominiums whose deed was not matched) are kept. Panel B divides the excess mass in $(X - 0.025X, X)$ and the missing mass in $(X, X + 0.05X)$ by the counterfactual density of sales per dollar at X , post minus pre; the buyer’s jump at \$1,000,000 is the mansion tax, unchanged in 2019. Supplemental Appendix Table A7 adds log changes in R and window sizes. Replication package v1.9, `referee_extensions.py` and `referee_v24.py`.

tax schedule is not proportional. House loan amounts are carried in the deposited linked panel. Condominium loan amounts come from a separate extract of the mortgages matched to condominium unit lots. That extract was not in the original replication deposit. The condominium row of Table 8 now uses the loans of the linked dataset (Loschi 2026a), which rebuilds them from public ACRIS records with a documented script: 47,702 condominium purchases whose loan matches on unit lot, sale date and price. The earlier extract, 47,610 loans, gave the same statistics at the reported precision. Section 5.2 uses the same loan-level

Table 7: What the 2019 notches cost in avoided tax

Threshold	Jump	Extra sales just	Missing minus	Tax avoided, \$M a year	
	at X	below X , a year	extra, a year	At the jump	0.05 X price cut
\$1,000,000 (control)	—	102 [17, 180]	−28 [−119, 64]	—	—
\$2,000,000	\$5,000	130 [103, 157]	−46 [−103, 17]	0.65 [0.52, 0.79]	1.05
\$3,000,000	\$15,000	54 [34, 72]	+17 [−29, 61]	0.80 [0.51, 1.09]	1.09
\$5,000,000	\$37,500	20 [6, 32]	+7 [−23, 41]	0.74 [0.22, 1.19]	0.96
Total, new thresholds				2.20 [1.63, 2.76]	3.10

Notes. Post-reform period, January 2020 to December 2025. Extra sales just below X are the excess mass in $(0.975X, X)$ after the reform minus the pre-reform excess rescaled to the post-reform counterfactual density; missing minus extra is the corresponding missing mass in $(X, 1.05X)$ less that excess. Estimator and windows as in Table 4; 95% bootstrap intervals in brackets. Each extra sale avoids at least the statutory jump at X (at \$3,000,000 the buyer’s 0.25 points and the seller’s 0.25-point base-rate rise). The tax figures assume that the extra sales below each threshold relocated from above it; the intervals cover sampling uncertainty conditional on the estimator and that assumption. The 0.05 X price-cut column is a scenario, not an identified bound: it assumes each extra sale also cut its price by the emptied range, 0.05 X , and adds every ad-valorem transfer tax on that reduction. The reform’s schedule (the supplemental tax and the base-rate rise) implies \$92.7 million a year on the 4,647 sales a year between \$2,000,000 and \$6,000,000. At the \$1,000,000 control, whose tax did not change, the extra sales are within the range of the never-taxed placebos. Replication package v1.9, `referee_v24.py`.

amounts.

Table 8: Gross schedule-implied borrower charge on observed loans, before credits, exemptions and CEMA adjustments

	Loans with an observed amount	Mean	Median	MRT as a share of price			
		LTV	LTV	Mean	Median	p10	p90
Houses	125,023	0.745	0.774	1.41%	1.44%	0.91%	1.86%
Condominiums	47,702	0.678	0.737	1.28%	1.37%	0.78%	1.60%

Notes. The borrower’s schedule, 1.80% below \$500,000 and 1.925% at or above, is applied to each loan at its own size rather than at a representative loan-to-value ratio, and expressed as a share of the recorded purchase price. The lower condominium benchmark accompanies a lower mean recorded loan-to-price ratio. The calculation does not identify why loan amounts or financing choices differ. A cash buyer pays none of it in either market, and a co-operative buyer pays none of it at all, since a share loan is personalty and outside Article 11. The schedule is applied to recorded principal; the calculation does not establish taxable new money and applies no credits or exemptions, such as the exclusion of the first \$10,000 of one- and two-family mortgages from the 0.30% component or the credit on first sales of condominium units (New York City Department of Finance 2025).

Consolidation agreements. A New York purchase can be financed by a Consolidation, Extension and Modification Agreement (CEMA), under which the buyer takes an assignment

of the seller’s existing mortgage and pays the recording tax only on new money. Where such agreements are used, recorded principal overstates the taxable amount. Retrieving the 29,511 M&CON documents (“MORTGAGE AND CONSOLIDATION” in the ACRIS code table) recorded since 2015 and matching them to parcels and dates, 56 of 72,830 purchases flagged cash, 0.08%, have a concurrent consolidation agreement. The consolidation screen concerns financing classification; it does not bound the adjustment from recorded principal to taxable new money among purchases already classified as financed.

5.2 Costing a notch-free change

Three statutory changes would narrow the difference in liability between a financed and a cash purchase: exempting owner-occupied residential purchase money from Article 11; imposing an equivalent charge on purchases that record no mortgage; or reassigning the borrower-borne components to the lender, as §253(1-a) already does for the special additional tax. Their economic incidence is not estimated here. The first forgoes revenue, including the §253(2) component that Tax Law §261 directs to the Metropolitan Transportation Authority, so it needs a replacement. The costing below covers all observed house and condominium purchases; owner occupancy is not identified.

The residential carve-out can be costed from the loans themselves. Summing the schedule over every financed purchase in the linked panel, each observed loan at its own size and with condominium loans taken from the linked dataset, gives **\$244 million a year** of gross schedule-implied borrower charge on residential purchase money in the four ACRIS boroughs between \$300,000 and \$6,000,000, of which \$38.5 million is the MTA-dedicated §253(2) component (Table 9, Panel A). Of the total, \$240.2 million rests on observed loan amounts and \$4.1 million on imputed ones, for the 3,470 houses and 370 condominiums without a usable loan amount. This is a bottom-up figure for the population the paper observes, not the city total: Staten Island and sales above \$6,000,000 lie outside it, consolidation-related adjustments are not identified, and co-operatives pay no recording tax. It is a fixed-sample gross schedule benchmark; net revenue forgone is not measured. Credits, exemptions and consolidation adjustments, including those affecting the \$38.5 million MTA component, would lower it.

Because notches move transactions (Section 4), the illustrations use continuous schedules. Table 9, Panel B, solves for three such schedules that raise the same \$244 million from the same purchases, paid by every house and condominium buyer regardless of payment method: 0.88% of the whole price; 1.56% of the price above \$500,000; or 1.31% between \$500,000 and \$2,000,000 and twice that above. Each is continuous in price: the marginal rate changes at

Table 9: The gross schedule-implied charge on residential purchase money, and three illustrative schedules matched to it

<i>Panel A. Gross schedule-implied borrower charge on purchase money, \$ million a year</i>				
Houses, loan amount observed (125,023 loans)	161.2			
Houses, loan amount imputed (3,470)	3.4			
Condominiums, loan amount observed (47,702 loans)	79.0			
Condominiums, loan amount imputed (370)	0.7			
Total	244.3			
of which the §253(2) MTA component	38.5			
Total on observed loan amounts only	240.2			
<i>Panel B. Smooth schedules on every purchase, same total</i>				
Schedule	Rate	Financed buyers paying less	Mean change per purchase Financed	Cash
Flat, on the whole price	0.88%	90.9%	−\$4,587	+\$11,120
On price above \$500,000	1.56%	89.9%	−\$5,045	+\$12,231
Two brackets, doubling at \$2,000,000	1.31%	91.0%	−\$5,335	+\$12,934

Notes. Merged house and condominium panel: 249,395 purchases between \$300,000 and \$6,000,000 in the four boroughs ACRIS covers, of which 176,565 financed. Staten Island sales, whose deeds and mortgages ACRIS does not record, are excluded. The charge is the borrower-side schedule applied to each observed loan’s recorded principal (1.80% below \$500,000, 1.925% at or above), before credits, exemptions and consolidation (CEMA) adjustments, so the totals are a fixed-sample gross benchmark, not net revenue. Condominium loan amounts come from the linked dataset (Loschi 2026a), matched to the panel on unit lot, sale date and price. Financed purchases without a usable loan amount carry the observed mean for their property type and price band. Sales whose unit, date and price match more than one record are left to imputation; the 47,702 condominium loans used here are those of Table 8. The two-bracket schedule charges the stated rate on price between \$500,000 and \$2,000,000 and twice it above. Replacement rates are solved so that each schedule raises Panel A’s total from the same fixed set of purchases; outside that set the schedules are not revenue-matched. Unfinanced purchases include any lender take-title transfers recorded as plain DEED (Section 3.2), which are not removed here. All three are continuous in price: marginal rates change at the breakpoints, and total liability does not jump. Figures are static: they hold transactions fixed and assign statutory incidence, so they are the size of the swap, not a forecast of its revenue.

a breakpoint, but total liability never jumps. Between 90% and 91% of financed buyers pay less, by about \$5,000 on average, and cash buyers pay about \$11,000 to \$13,000 more. Keyed on property class, the flat rate would be 1.02% for houses and 0.68% for condominiums. A schedule starting at \$1,000,000, the threshold of the cash-purchase tax proposed in 2026, would miss much of the purchase population that now pays the recording tax. The schedules show the size of the swap. They hold transactions fixed and assign statutory incidence, and nothing in Section 4 identifies how buyers would respond to a continuous schedule.

6 What this paper cannot settle

The statutory measurement is a gross benchmark. The charge measured in Section 5.1 applies the borrower-side schedule to recorded principal on purchase-money mortgages, before credits, exemptions and consolidation adjustments. The consolidation screen concerns financing classification; it does not bound the adjustment from recorded principal to taxable new money among purchases already classified as financed. Neither figure includes recording, title or legal costs unrelated to the tax itself.

No elasticity, so no optimal rate. This paper measures the size of the statutory asymmetry and documents how a notch-shaped instrument behaves; it does not determine the rate at which any corrective charge would maximise welfare. That requires knowing how many transactions would relocate or fail to occur at a given rate, and nothing here identifies that. The notch experiment measures responses to discrete jumps and does not bound or identify responses to a continuous schedule. A full welfare analysis would also account for the foregone revenue and the economic incidence of each instrument, which this paper does not estimate.

The revenue replacement is costed only within the sample. Section 5.2 costs the purchase-money component bottom-up at \$244 million a year for the four ACRIS boroughs between \$300,000 and \$6,000,000, with 98% of it resting on observed loan amounts. A legislature would need the same figure for the whole city, including Staten Island and higher-priced sales, net of credits, exemptions and consolidation adjustments, and an estimate of how a replacement schedule would change transaction volume. None of that is available from the public data used here.

Foreclosure transfers are not removed. Referee’s deeds and deeds in lieu of foreclosure are recorded as plain DEED, and a companion paper reports that 94–95% of those recorded in 2016–2025 appear in the Department of Finance sales file at their recorded amount (Loschi 2026b). Their consideration can follow a legal rule rather than a negotiated price, and a lender taking title records no purchase mortgage. Such transfers therefore remain in this paper’s price sample and among the unfinanced purchases of Table 9. They do not enter the loan-level burden of Table 8, which uses financed purchases only. The companion paper reports that the recorded amounts of these transfers are rarely round and, over 2003–2025, have medians below \$500,000, far from the 2019 thresholds. Section 4.10 drops them from the threshold windows using the companion classifier and finds no change, but that classifier

has been checked only against AI readings of sampled deed documents, not against a verified record.

The market-level question is left open. Whether a charge on cash purchases is justified beyond the statutory asymmetry measured here depends on how much of the financing-associated price gap reflects execution risk that a seller is rationally compensated for, as against liquidity, unpriced advantage, changing property condition, or selection into cash purchases. A companion paper takes up that question on the same administrative data (Loschi 2026c). It finds a positive repeat-sales association for houses that depends heavily on who pays cash. It is small when the cash buyer is a company and falls by almost half when pairs with a same-surname deed are excluded, and the condominium contrasts are close to zero; its design does not separate the candidate mechanisms. Nothing in this paper’s statutory measurement, threshold evidence or static costing depends on how that question is eventually resolved, and nothing here should be read as adjudicating it.

7 Conclusion

New York’s 2019 reform allows a before-and-after comparison at the same transfer-tax thresholds. At each well-populated new notch, sales moved to just below the threshold. Each shift is larger than at any of sixteen never-taxed round prices below \$6,000,000 and starts with the reform, while the always-taxed \$1,000,000 threshold changes little. The shift appears in houses and condominiums, in Manhattan and in the other boroughs, and in sales without a company seller. Removing foreclosure and lender deeds does not change it. At \$2,000,000 and \$3,000,000 it empties a range of prices two to three times as wide as the tax jump. If the extra sales below each threshold came from above it, the avoided tax is \$2–3 million a year, 2–3% of the revenue the reform’s schedule implies between \$2,000,000 and \$6,000,000.

The point estimates at \$2,000,000 and \$5,000,000 are nearly equal, although the buyer’s dollar jump is 7.5 times larger at \$5,000,000. The sampling bootstrap rejects a response proportional to the buyer’s rate ($p = 0.008$); placebo comparisons give weaker, reference-dependent evidence. The two thresholds serve different markets, so the comparison does not vary the rate alone, and the difference between them is imprecise. The design does not identify the effect of changing the statutory payer or of a continuous schedule. The rejection of proportionality is a comparison of estimated responses at different price levels, conditional on the fitted counterfactual; it is not a causal estimate of a rate-response function. For the design of transaction taxes, the finding is that even a quarter-point notch moves transactions, and within the observed range a lower rate did not bring a proportionally smaller response.

The same question of shape applies to New York City’s mortgage recording tax. Applied to recorded principal, the borrower-side schedule implies a gross charge that averages 1.41% of price on 125,023 observed house loans and 1.28% on 47,702 observed condominium loans, before credits, exemptions and consolidation adjustments. The paper does not observe tax payments or who ultimately bears the tax. For purchases between \$300,000 and \$6,000,000 in the four ACRIS boroughs, the fixed-transaction gross total is \$244.3 million a year, with observed loan amounts for all but 2% of financed purchases (Table 9). For example, a smooth schedule of 0.88% of price matches this amount arithmetically in the same sample. What such a schedule would raise after a policy change, and who would bear it, are open questions.

Declarations

Data availability. All data are public and free. Sales are New York City Department of Finance annualized calendar sales, NYC Open Data resource `w2pb-icbu`. Deeds and mortgages are ACRIS Real Property Master and Legals, resources `bnx9-e6tj` and `8h5j-fqxa`, linked to the sales in a published dataset (Loschi 2026a), concept DOI `10.5281/zenodo.22977532`. The replication package, concept DOI `10.5281/zenodo.22880953`, holds the analysis data and the code for every table and figure; version 1.9 reproduces this version without network access. Its scripts for the original ACRIS linkage are included for provenance, and the linked dataset rebuilds the same links from the public sources with its own code. Table 8 and the costing in Table 9 use the condominium loans of the linked dataset (47,702 matched loans); an earlier, undeposited extract of 47,610 loans gave the same Table 8 statistics at the reported precision. Preprint versions are under concept DOI `10.5281/zenodo.22925301`. The companion papers’ replication packages cover their own analyses (Loschi 2026c;b); no result in this paper is taken from them.

Declaration of generative AI use. Anthropic’s Claude and OpenAI’s ChatGPT, including Codex, were used for coding, analysis scripts, reference formatting, drafting and editing, and to cross-check each other’s analyses, in earlier drafts and in every revision through version 2.6. The author checked the reported results against the deposited outputs and is responsible for the submitted version.

Declaration of competing interest. The author founded HolaCasaClara, a free comparator of Argentine mortgage loans, which has no connection to New York housing records or this study. No other competing interests are reported.

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